

GovOS

Complete Frontend Development Plan

CivicPath Public Experience • Government Operations • Constituency Progress •
Role-Based Dashboards

Core Vision Together, We Build Better Communities.	Product Direction From citizen voice to visible governance.
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Theme: Modern Civic Technology • Positive Governance • Transparency • Community
Progress

1. Project Understanding

GovOS is a Multi-Tenant Government Operating System designed to improve how citizens, government departments, officers, administrators, and constituency representatives interact.

The public-facing citizen experience is branded as **CivicPath — Digital Governance**. The platform must not feel like a traditional government complaint portal.

Central Philosophy

FROM: Citizens reporting problems.

TO: Citizens participating in visible community improvement.

Core Governance Journey

Citizen Voice → Issue Reported → Validation → Department Assignment → Officer Action → Work Completed → Evidence Submitted → Verification → Visible Community Progress

GovOS is therefore not just a complaint management system. It is a governance operations and public progress platform.

2. Product & Experience Principle

The public experience must avoid a negative, failure-focused narrative.

Do Not Make the Platform Feel Like

- A government failure tracker
- A complaint database
- A political campaign website
- A system focused only on broken infrastructure and unresolved problems

The Experience Should Communicate

- Community improvement
- Transparency
- Visible progress
- Government accountability
- Work completion
- Citizen participation
- Positive civic engagement

Preferred language includes: **Report an Issue, Community Progress, Recently Resolved, Improvements in Your Area, Work in Progress, Building Better Communities, Visible Governance.**

3. Existing Repository Rules

- Inspect the complete repository before implementation.
- Identify existing frontend applications, routes, components and reusable code.
- Preserve useful working implementation.
- Identify duplicate, dead and broken code before cleanup.

- Do not blindly rewrite the project.
- Do not invent backend APIs.
- Do not scatter fake production data across UI components.

4. Frontend Application Structure

A. Public Civic Platform

Home • Report Issue • Track Issue • Public Dashboard • Constituency Dashboard • Recently Resolved • How It Works • Resources

B. Authenticated GovOS Platform

Login • Authentication • Dashboard Router

C. Role-Based Dashboards

SUPER_ADMIN • TENANT_ADMIN • DEPARTMENT_HEAD • OFFICER • REPRESENTATIVE / MLA • CITIZEN

Public and authenticated layouts must remain clearly separated.

5. Design System & Visual Theme

The visual direction is **Modern Civic Technology**: clean, trustworthy, optimistic, transparent, professional and human-centered.

- Clean layouts and generous whitespace
- Soft neutral backgrounds
- Modern reusable cards
- Subtle shadows and rounded corners
- Strong typography hierarchy
- Accessible contrast
- Minimal gradients
- Calm, purposeful animation

Semantic Colour Direction

- **Deep Governance Navy**: trust, navigation and institutional structure
- **Community Green**: completion, progress and positive improvement
- **Progress Blue**: active workflows and information
- **Warm Orange**: actions and important highlights
- **Red**: genuine critical alerts only

6. Global Component System

- AppShell, PublicNavbar, DashboardSidebar, MobileNavigation and Footer
- PageHeader, SectionHeader and KPIGrid
- StatCard, ProgressCard, StatusBadge, DepartmentBadge and WardBadge
- DataTable, FilterBar, SearchInput and Pagination
- EmptyState, LoadingState and ErrorState

- ConfirmationDialog, Modal, Drawer and Toast Notifications
- Tabs, Timeline, ActivityFeed and EvidenceViewer
- MapContainer abstraction

Avoid duplicated one-off UI. Components must be reusable and consistent.

7. CivicPath Public Website

7.1 Home Page

Hero: “Together, We Build Better Communities.”

Supporting message: “Report issues, track progress, and see how your community is improving.”

Primary CTA: Report an Issue **Secondary CTA:** Track Progress

Hero Metrics: Citizens Connected • Issues Resolved • Departments Connected • Satisfaction Rate

Visuals should show a positive community: citizens, families, green spaces, clean streets, public infrastructure and a growing city.

7.2 How It Works

- **01 — Report Your Issue:** Share details, photos and location.
- **02 — Validation & Routing:** The issue is reviewed and sent to the right department.
- **03 — Expert Action:** The responsible team begins work with clear accountability.
- **04 — Resolution & Updates:** Track progress and see verified improvements.

7.3 Community Progress

Section title: “See the Difference We’re Making”

- Recently Resolved
- Community Improvements
- Active Government Work
- Department Performance

Examples: Pothole repaired • Water supply restored • Street lights replaced • Sanitation improved

7.4 Positive CTA

Heading: “See Something That Needs Attention?”

Report it today and help make your city better for everyone.

7.5 Footer

CivicPath • Quick Links • Resources • FAQs • Privacy Policy • Terms of Service • Helpline • Appropriate GovOS/company branding based on repository requirements.

8. Report Issue Flow

Step 1 — Issue Details

Issue title • Detailed description • Category • Sub-category • Citizen-perceived urgency

Important: citizen urgency should not automatically become the official government severity.

Step 2 — Location

Address • Ward • Constituency • Map location • Optional GPS

Step 3 — Attachments

Photos and supporting documents.

Step 4 — Contact / Authentication

Mobile number and OTP flow.

Step 5 — Review & Submit

Show issue summary, location, attachments and privacy/consent acknowledgement.

Success message: “Thank You for Helping Improve Your Community”

Timeline: Submitted → Under Review → Assigned → Work Started → Completed → Verified

9. Track Issue / Track Community Progress

The page should not feel like a cold complaint lookup system.

Title: “Track Community Progress”

Search by Issue ID or registered mobile number.

Public Progress Timeline

Submitted → Validated → Assigned → Work Started → Work Completed → Verification → Verified Completed

- Department
- Current status
- Estimated timeline where available
- Progress updates
- Completion evidence
- Verification status

10. Public Dashboard

Title: “Community Progress Dashboard”

Show public aggregated data only.

KPIs

- Total Service Requests
- Issues Resolved
- Work in Progress
- Average Resolution Time

Visual Analytics

- Category Breakdown
- Monthly Improvement Trend

- Resolution Performance
- Department Contribution
- Recent anonymized activity

Privacy Rule: Never expose citizen personal data, phone numbers, private addresses or sensitive issue information.

11. Constituency Public Dashboard

This must be a **Governance Progress Page**, not an MLA promotional website.

Example: “Dharwad Constituency — Community Progress & Public Services”

11.1 Constituency Overview

Constituency name • Wards covered • Departments connected • Public service metrics

11.2 Community Progress

Issues resolved • Work completed • Active improvements • Average response time

11.3 Improvements Across the Constituency

Roads • Water • Sanitation • Electricity • Infrastructure • Environment

11.4 Ward Performance

Public aggregated metrics only.

11.5 Government Work in Progress

Work name • Department • Area • Progress percentage • Expected completion where publicly appropriate

11.6 Community Map

Show aggregated completed improvements, active work and department coverage. Never expose private citizen issue locations.

12. Representative / MLA Dashboard

This is **private and authenticated**. The representative is an **oversight role**, not the person manually solving every issue.

Capabilities

- Constituency overview
- Ward performance
- Department performance
- SLA breaches
- Critical unresolved issues
- Major infrastructure projects
- Escalations
- Directives

Representative Actions

- Raise directive
- Request status
- Escalate issue
- Review department performance

13. Super Admin Dashboard

- Tenant overview and provisioning
- Platform health
- User administration
- Role management
- Global analytics
- Audit visibility

System-level UI must clearly distinguish global administration from tenant data.

14. Tenant Admin Dashboard

- Tenant overview
- Departments and officers
- Wards
- Service categories
- SLA configuration
- Users
- Analytics and reports

15. Department Head Dashboard

- Department workload
- Assigned work
- Team performance
- SLA status
- Escalations
- Assets
- Projects
- Reports

16. Officer Dashboard

Main screen: My Work

- Assigned tasks
- Priority

- SLA deadline
- Location
- Current status

Officer Actions

Accept → Start Work → Add Update → Upload Evidence → Mark Work Completed

Critical workflow rule: Marking work completed must not automatically mean final closure.

Correct lifecycle: **Work Completed** → **Evidence Submitted** → **Quality / Verification** → **Verified Completed**

17. Citizen Dashboard

- My Reports
- Track Progress
- Updates
- Recently Completed
- Community Progress

Keep the experience simple, mobile-first and easy to understand.

18. Status System

Use a consistent public-facing lifecycle:

SUBMITTED → **UNDER REVIEW** → **ASSIGNED** → **WORK IN PROGRESS** → **WORK COMPLETED** → **UNDER VERIFICATION** → **VERIFIED COMPLETED** → **CLOSED**

Preserve backend compatibility. Do not silently change backend enum values; use a frontend mapping layer where necessary.

19. Multi-Tenant Frontend Rules

- The frontend must be tenant-aware.
- Never allow users to manually choose arbitrary tenant IDs.
- Tenant context must come from authentication, session or domain configuration.
- Never trust `tenant_id` from a form field.
- Frontend route guards are UX controls only; backend remains the security authority.
- Support future tenant branding: logo, organization name, accent and domain/subdomain.

20. Security Requirements

- Never hardcode JWT secrets, API credentials or other secrets.
- Use environment variables.
- Centralize authentication and API handling.
- Handle token expiration, unauthorized and forbidden states correctly.
- Provide secure logout.
- Validate inputs and escape output appropriately.

- Do not expose tenant authorization logic or citizen data publicly.

21. API Architecture

Create a clean centralized API abstraction.

- auth.service
- complaint/service-request service
- dashboard.service
- department.service
- officer.service
- project.service
- asset.service
- constituency.service

Do not place random API calls throughout UI components.

If backend APIs are not ready, use a clean adapter layer such as `API_MODE=mock` / `API_MODE=production`. Do not scatter mock data.

22. State Management

Server State: Use React Query or the existing repository standard.

Client State: Use Zustand or the existing project state system.

Do not unnecessarily duplicate server data into global client state.

23. Responsive Design

- Mobile
- Tablet
- Laptop
- Desktop

Citizen experience must be mobile-first. Government dashboards should support desktop operational workflows while remaining usable on tablets.

24. Accessibility

- Keyboard navigation
- Visible focus states
- Accessible labels
- Semantic HTML
- Sufficient contrast
- Screen-reader support
- Error messages correctly associated with inputs

25. Performance

- Code splitting
- Lazy-loaded routes
- Image optimization
- Bundle-size discipline
- Efficient dashboard rendering
- Large table virtualization where needed

Do not introduce heavy libraries without a clear requirement.

26. Animation

Use subtle professional motion:

- Page transitions
- Card hover
- Progress animation
- Timeline transitions
- Status updates

Avoid excessive or distracting motion. Respect reduced-motion preferences.

27. Scalable File Structure

src/

app/ • routes/ • layouts/

components/ui/ • components/common/ • components/dashboard/ • components/public/

features/auth/ • features/issues/ • features/departments/ • features/officers/ • features/projects/ • features/assets/ •

features/constituency/

services/ • hooks/ • store/ • types/ • utils/ • constants/

28. Implementation Process

F0 — Repository Audit & Cleanup

- Existing frontend inventory
- Existing routes
- Reusable components
- Duplicate code
- Broken code
- Recommended cleanup plan

F1 — Design System

Colors • Typography • Spacing • Buttons • Inputs • Cards • Status badges

F2 — CivicPath Public Experience

Home • Report Issue • Track Issue • Public Dashboard

F3 — Constituency Public Dashboard

Constituency overview • Community progress • Ward performance • Work in progress

F4 — Authentication

Login • OTP flow • Session handling • Role routing

F5 — Government Role Dashboards

Super Admin • Tenant Admin • Department Head • Officer • Representative

F6 — Integration Layer

Connect to verified backend API contracts.

F7 — Quality Assurance

Routes • Responsive design • Role routing • Loading/error/empty states • Accessibility • Build verification

29. Quality Rules

- Compile successfully
- Pass linting where configured
- Avoid TypeScript errors
- Avoid unused imports
- Avoid console errors
- Avoid duplicated components
- Avoid dead code
- Use consistent naming
- Implement loading, error and empty states
- Do not mark work complete without verification

30. Final Deliverables

- Frontend Architecture Summary
- Route Map
- Component Inventory
- Role Dashboard Matrix
- Public vs Private Page Matrix
- API Integration Matrix
- Responsive Verification
- Security Review
- Remaining Backend Dependencies
- Build Verification Result

Execution Priority

M0.5 / Foundation Cleanup → F0 Frontend Audit → F1 Design System → F2 Public CivicPath → F3 Constituency Dashboard → F4 Authentication → F5 Role Dashboards → F6 Backend Integration → F7 Production QA

This sequence prevents building a visually polished frontend on top of unstable architecture.

Final Product Standard

GovOS must look and feel like a serious, modern civic technology platform built around **Transparency, Progress, Community, Accountability and Improvement**.

“Together, We Build Better Communities.”

Citizen Voice → Government Action → Verified Work → Visible Governance.